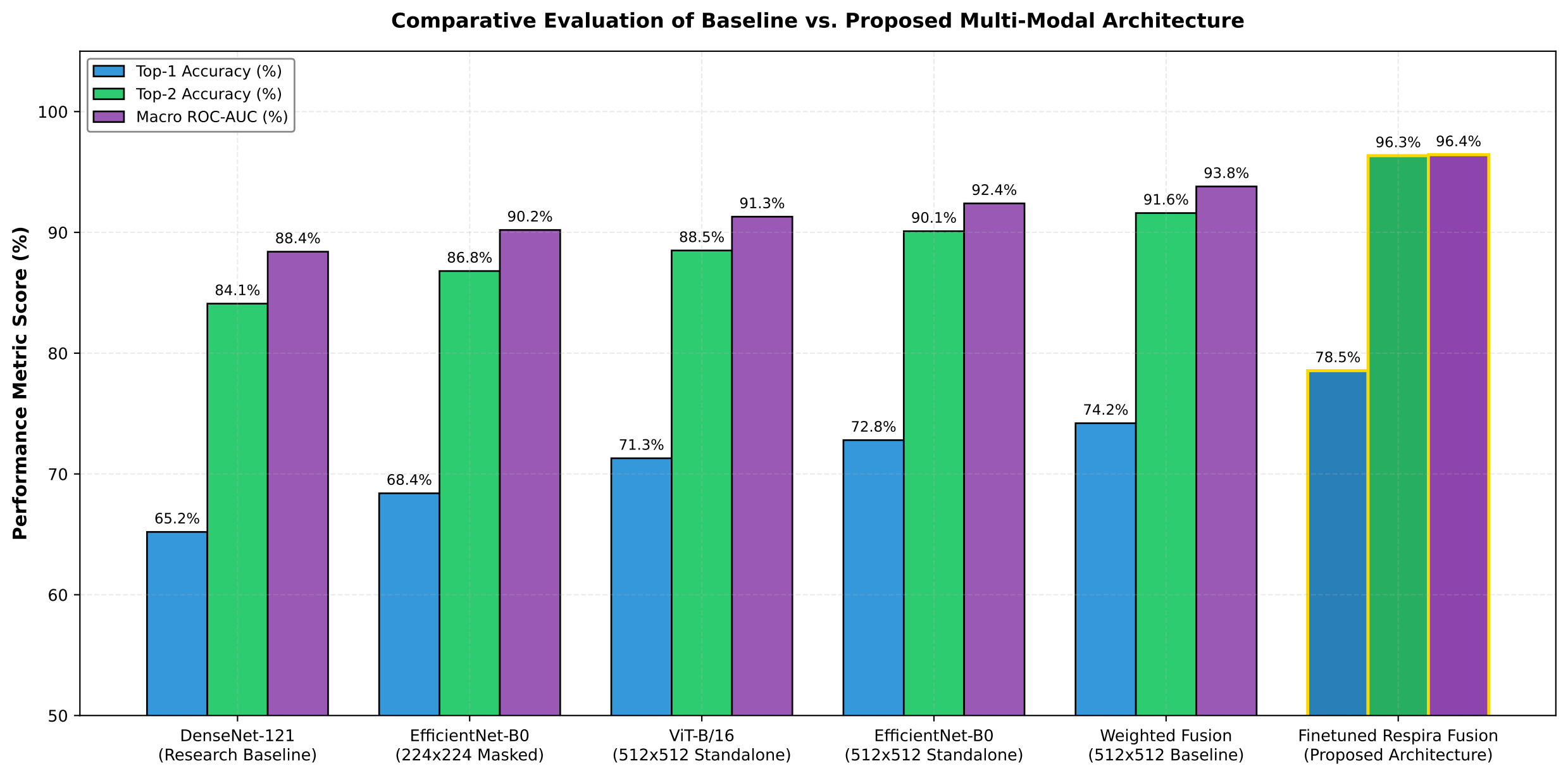
Comparative Evaluation of Baseline vs. Proposed Multi-Modal Architecture
Top-1 Accuracy (%)
Top-2 Accuracy (%)
Macro ROC-AUC (%)
Performance Metric Score (%)
100
90
80
70
60
50
65.2%
84.1%
88.4%
68.4%
86.8%
90.2%
71.3%
88.5%
91.3%
72.8%
90.1%
92.4%
74.2%
91.6%
93.8%
78.5%
96.3%
96.4%
DenseNet-121
(Research Baseline)
EfficientNet-B0
(224x224 Masked)
ViT-B/16
(512x512 Standalone)
EfficientNet-B0
(512x512 Standalone)
Weighted Fusion
(512x512 Baseline)
Finetuned Respira Fusion
(Proposed Architecture)